Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Full-stack AI engineer - built AgentProbe **end to end**: LLM agents (LangGraph), RAG pipeline (ChromaDB), structured extraction, LLM Router for multi-provider inference, and a 5-metric eval engine. 8+ years production backend engineering at scale (1M+ records, 100K+ MAU, Python/TypeScript/AWS). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

End-to-End AI Delivery | LLM Agents / LangGraph | RAG Pipelines / ChromaDB | Structured Extraction | LLM Inference & Routing | LLM Evaluation & Judges | Python / TypeScript / FastAPI | AWS / Docker / Microservices | PostgreSQL / SQL Tuning | NLP / Intent Modeling | Production Debugging | Full-Stack (React + Backend)

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **20+ full-stack production modules** (Python, TypeScript, PostgreSQL) including billing automation, payment pipelines, and decision support workflows - delivering features end to end across **120+ Agile sprints** for **100K+ monthly active users**.
- Cut microservices execution time by **50%** via Docker-based containerization; built distributed AWS pipelines (S3, Glue, EC2) processing **1M+ records** with automated quality checks and zero security incidents.
- Improved production query performance by **30%** by architecting PostgreSQL schemas, data access layers, and indexing strategies - maintaining **99.5% uptime** and sub-10 bug count for 6+ consecutive months.
- Eliminated **80% of manual operational workload** via Python and PL-SQL automation; reduced deployment times by **40%** with TDD/BDD quality gates.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted Python and database mentoring for 30+ undergraduates through sprint-style feedback cycles.

PROJECTS

AgentProbe: Full-Stack AI System - Agents, RAG, Eval OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure · TypeScript

Built **end to end** - zero team, zero cloud budget. Architecture: **LLM Router** for dynamic multi-provider inference routing (Ollama, OpenAI, Anthropic, Groq); 3 async LangGraph agents (Orchestrator, RAG Tracer, Auto Fix); **ChromaDB RAG pipeline** with structured retrieval evaluation; **structured extraction** layer mapping queries to eval schemas; **5-metric LLM judge engine** (correctness, hallucination, retrieval, completeness, compliance); **Auto Fix agent** for automated prompt iteration on failure; **15-endpoint FastAPI backend**; React observability dashboard. 100% local at zero cost. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Structured Extraction for Medical Records

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

Structured extraction pipeline: natural language intent (EN/ES) → schema extraction via LLMs + BERT + spaCy → structured FHIR API call. HIPAA-compliant RBAC, containerized. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive Modeling on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Feature engineering → model development → evaluation → deployed dashboard. **85%+ accuracy**; outputs designed for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, LLM Inference & Routing, RAG, ChromaDB, Structured Extraction, LLM Judges, Prompt Engineering, Ollama, OpenAI, Anthropic, Groq, FastAPI

ML Stack: PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, Pandas, BERT, spaCy, Feature Engineering, Predictive Modeling

Databases: PostgreSQL, MySQL, Oracle, NoSQL, GraphQL - schema design, query tuning, indexing

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices

Testing: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum